Representation Learning in Heterogeneous Graphs: A Relation-Aware Homogenization Framework

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***Abstract*—Heterogeneous graphs are widely used these days to model complex real-world systems, but their different node and relation types make representation learning challenging. Existing heterogeneous graph neural networks heavily relies on predefined meta-paths or relation-specific architectures, which can make them less flexible across different tasks. In this work, we propose a relation-aware heterogeneous graph homogenization framework that handles heterogeneity during representation learning and produces a unified homogeneous embedding space.Firstly, a type-specific projection module maps features from different node types into a shared latent space. Next, we use a bipartite projection to create same-type connections based on shared neighbors. For each relation, a Learnable Polynomial Aggregation module learns how much information to aggregate from nearby and distant neighbors, allowing the model to adapt to different neighborhood ranges. The representations learned from different relations are then combined through adaptive relation fusion, which learns the contribution of each relation. Finally, a residual connection preserves the original node information and helps reduce over-smoothing.Experiments on standard heterogeneous graph benchmarks show that the proposed framework performs competitively with specialized heterogeneous graph neural networks and conventional homogeneous baselines. Our preservation analysis further evaluates whether important semantic, relational, structural, and meta-path information is retained after homogenization. Overall, the framework provides a simple way to handle heterogeneous graphs by addressing heterogeneity during representation learning, allowing standard homogeneous graph learning methods to be applied without modifying their architectures.**

***Index Terms*—Heterogeneous Graphs, Graph Neural Networks, Graph Homogenization, Bipartite Projection, Polynomial Aggregation, Representation Learning.**

I. Introduction

GRAPH Neural Networks (GNNs) emerged in recent years as one of the most powerful approches for graph representation [1]–[3]. GNNs learn embeddings via By iteratively aggregating information from neighbouring nodes that capture both the structural characteristics of a graph and association with entities, due to their expressive power. GNNs have achieved remarkable success across a broad range of applications in this decade, including social network analysis [4], protein structure prediction [5], traffic forecasting [6], and drug discovery [7]. In addition ,as an area of interest for machine learning on relational data , GNN-based methods demonstrate strong performance for several downstream tasks like node classification, link prediction, graph classification,node clustering,graph clustering and other representation learning tasks [8], [9].

Most GNN architectures in litratures designed for homogeneous graphs, considering all nodes and edges are assumed to have to a single type and share a common representation space [10]. This assumption simplifies representation learning, but real-world systems the representation is complex and messy . Many domains naturally generate heterogeneous graphs incorporating multiple types of entities connected through multiple types of relationships [11] in practice. As an example, an academic network simultaneously contains authors, papers, venues, and research topics linked through authorship, citation, publication, and topical associations [12] on the other hand biomedical knowledge graphs contains genes, proteins, diseases, drugs, and biological pathways through a complex and rich collection of interaction types [13]. The semantic diversity and richness lying inside such networks provides valuable information that is often lost when heterogeneous structures are naively transformed into homogeneous ones [14]. Consequently, heterogeneous information networks (HINs) encode significantly richer information than homogeneous graphs and therefore they require learning frameworks capable of modeling both structural and semantic heterogeneity effectively [11], [15].

Limitations of Existing Approaches

Research on heterogeneous GNNs has progressed along two important categories. The first category of approaches constructs *meta-paths*, composite relational sequences that connect nodes of the target type through intermediate entities. Work such as HAN [12] and MAGNN [16] constructs separate attention based aggregators along each meta-path, then fuses the resulting representations hierarchically. They are elegant in both theory and design but meta-path-based methods carry a critical dependence on domain expertise: the selection of semantically meaningful meta-paths requires prior knowledge of the graph schema, and suboptimal selections degrade performance substantially. Attempts to automate meta-path discovery, as in GTN [17] and DiffMG [18], alleviate this burden at the cost of significant computational overhead. Furthermore, because meta-paths are topologically shallow by construction, they capture only limited orders of relational context.

The second category avoids meta-paths entirely, instead modeling heterogeneity through relation-specific parameter matrices or attention mechanisms. R-GCN [10] maintains separate convolutional weights for each relation type; HGT [19] employs a per-relation transformer; R-HGNN [20] learns explicit relation representations that modulate node updates. These methods are more flexible in their structural assumptions but come with a parameter count that scales linearly with the number of relation types, rendering them inefficient on graphs with rich relational vocabularies. More broadly, both families of methods couple the handling of heterogeneity tightly to the downstream model architecture, foregoing the possibility of re-using advances in homogeneous GNN design.

A recent empirical study [21] has drawn attention to a frequently overlooked observation: carefully initialized homogeneous GNNs can approach, and occasionally match, the performance of heavily engineered heterogeneous counterparts on certain datasets. This suggests that the heterogeneity of a graph need not require a correspondingly heterogeneous architecture, provided that the graph representation itself absorbs the relational semantics.

Figure 1 illustrates the four principal strategies through which prior work has approached the heterogeneous graph learning problem, contrasted against our proposed homogenization view. Panel (a) depicts the original heterogeneous graph, where nodes of multiple types coexist alongside typed edges that carry distinct relational semantics. Panel (b) shows the meta-path paradigm: the heterogeneous graph is decomposed into a collection of homogeneous subgraphs, one per meta-path, each encoding a particular composite relation between nodes of the target type. While this decomposition makes the graph amenable to standard GNN processing, it requires prior specification of semantically meaningful relational sequences and discards all information that does not align with the chosen meta-paths. Panel (c) illustrates the relation-based approach, in which separate subgraphs are maintained for each relation type and processed by relation-specific modules; the complexity of such architectures scales proportionally with the number of distinct relations. Panel (d) depicts the homogeneous graph with relation-specific weights, where heterogeneity is encoded through learned scalar importances assigned to each relation. Our framework, by contrast, achieves a principled and expressive form of homogenization that incorporates bipartite correction, multi-hop spectral diffusion, and adaptive relation fusion, yielding a latent representation that is simultaneously structure-aware, relation-aware, and directly compatible with any standard homogeneous GNN.

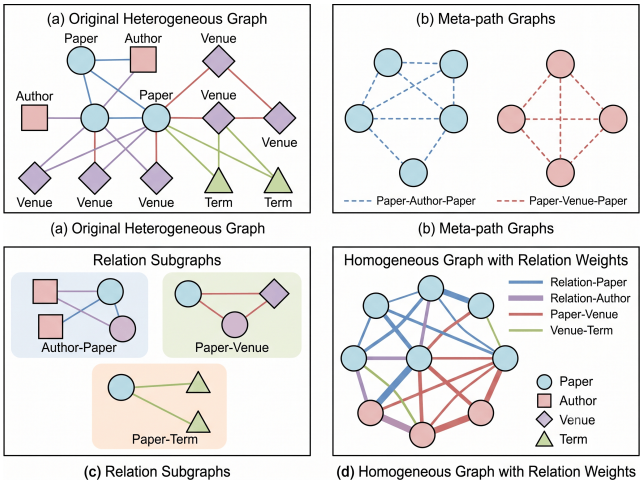


Fig. 1. Four representative paradigms for handling heterogeneous graphs, illustrated on an academic relational network. **(a)** The original heterogeneous graph, containing nodes of multiple types (papers, authors, venues, terms) and typed relational edges. **(b)** Meta-path-based homogeneous graphs: the heterogeneous structure is decomposed into separate homogeneous graphs, each defined by a composite relational sequence such as paper-author-paper or paper-venue-paper. **(c)** Relation-based subgraphs: the graph is partitioned into relation-specific subgraphs, processed by dedicated modules. **(d)** Homogeneous graph with relation-specific weights: heterogeneity is absorbed into a single weighted homogeneous graph, where each relation contributes with a learned scalar importance. Our proposed framework extends paradigm (d) through differentiable bipartite correction, relation-aware Learnable Polynomial Aggregation, and adaptive semantic fusion, achieving richer homogenization without sacrificing relational identity.

A Homogenization Perspective

This observation motivates us to change our perspective. Rather than asking how to adapt a GNN architecture to accommodate heterogeneous structure, we ask a different question: how can heterogeneous graph structure to accomodate into a single homogeneous latent space, such that any standard homogeneous GNN can be applied directly and effectively without explicitly changing? We call this approach *heterogeneous graph homogenization*, and we argue that it offers a principled and practically better alternative to relation-specific architectural design.

As we already discussed despite the rich semantics of heterogeneous graphs, learning effectively from such structures remains a challenging problem [14], [15]. This difficulty arises from the heterogeneity of both entities and relations. First, different node types are often associated with features originating from different semantic and statistical spaces, making direct comparison and aggregation nontrivial [22]. Consequently, an effective learning framework must create a common representation space while preserving type-specific information. Second, heterogeneous graphs frequently contain multiple cross-type interactions and bipartite relational structures, where edges connect fundamentally different categories of entities. Applying conventional message-passing mechanisms directly to such graphs can harm the semantic information and also can degrade representation quality [10]. Third, the presence of multiple relation types changes the lavel of importance, redundancy, and noise across interactions. Since not all relations contribute equally to downstream tasks, principled mechanisms for relation-aware aggregation and information fusion are essential [3], [12].Addressing these challenges motivates us for the development of a homogenization framework that transforms heterogeneous graphs into a unified homogeneous representation which preseves as well as respects their structural and semantic characteristics .

Proposed Framework

We propose a relation-aware heterogeneous graph homogenization framework that addresses each of these challenges through a step by step processing stages. We begin by projecting the features of each node type into a common latent geometry via type-specific linear projection, which primarily eliminates dimensional and semantic incompatibility across node types. We then normalize each relation-specific adjacency matrix to make an stable propagation operators.

Our central technical contribution is a *bipartite correction* mechanism. For relations that connect nodes of distinct types, we observe that direct application of the propagation operator would transmit representations across type boundaries, corrupting the homogeneous structure of the embedding space. We resolve this by replacing the bipartite operator with its second-order composition, $\tilde{P}_r = P_r P_r^\top$, which defines a same-type similarity graph: two source nodes become connected if and only if they share common target neighbors in the original relation. This construction recovers the essential semantics of meta-path expansion in a fully differentiable, matrix-based form, without requiring any manual schema specification.

Building on the corrected operators, we applied relation-specific Learnable Polynomial Aggregation to compute multi-hop embeddings for each relation. The aggregator weights are learned via a softmax parameters, allowing the model to balance local and global neighborhood context in a data-driven mechanism. This process can also be interpreted as a learnable spectral filter applied independently to each relation graph, establishing connection to spectral graph theory. Finally, by fusing the relation-specific embeddings via adaptive weighting, and stabilize the final representation through a residual connection with the original projected features.

The output of our framework is a single matrix $Z_{\text{final}} \in \mathbb{R}^{N \times d}$ that encodes the original relational semantics in a unified homogeneous latent space. This representation can be used by by any standard homogeneous GNN or downstream machine learning model, while reducing the complexity of the end-to-end pipeline.

Summary of Contributions

Our main contributions are as follows:

- We propose a unified heterogeneous graph homogenization framework that transforms heterogeneous relational structures into homogeneous graph representations compatible with standard GNN backbones, shifting the complexity from the model architecture to the graph representation.
- We introduce a differentiable bipartite correction mechanism that transforms cross-type bipartite relations into same-type similarity graphs through a second-order matrix transformation, providing a differentiable and relation-aware generalization of meta-path-based graph construction.
- We develop a relation-specific Learnable Polynomial Aggregation scheme with learnable hop-wise weights, which enables data-driven spectral filtering that captures multi-hop neighborhood structure independently for each relation type.
- We design an adaptive relation-wise fusion layer that integrates information from multiple relations into a unified semantic representation while reducing the influence of noisy relations.
- We demonstrate the effectiveness of the proposed framework through extensive experiments on standard heterogeneous graph benchmarks, showing that it is both competitive with specialized heterogeneous GNNs and directly applicable to standard homogeneous GNN backbones.

The rest of this paper is organized as follows. Section II reviews related work. Section III establishes formal notation and background. Section IV describes the proposed framework in detail. Section V provides theoretical analysis. Section VI introduces the homogenization preservation analysis. Section VII reports experimental results. Section VIII concludes.

II. Related Work

A. Graph Neural Networks

Graph Neural Networks(GNNs) provides the dominant approch in litrature for learning on graph-structured data . Spectral methods define convolution operations through the graph Laplacian eigenbasis [23], [24], while spatial methods directly aggregate messages from local neighborhoods [2], [25]. GCN [1] linearizes spectral convolution and achieves strong empirical performance with a simple layer-wise propagation rule. GAT [3] introduces attention-weighted neighborhood aggregation, enabling the model to assign differential importance to neighboring nodes. GIN [26] connects message passing to the Weisfeiler-Lehman graph isomorphism test and establishes tight expressive power bounds. MixHop [27] and SGC [28] demonstrate that multi-hop neighborhood information can be effectively incorporated through polynomial filter approximations. All of these methods, however, are designed for homogeneous graphs and do not directly generalize to heterogeneous settings.

B. Heterogeneous Graph Neural Networks

a) Meta-path-based Methods.: HAN [12] introduces the meta-path based methodology for heterogeneous graph learning, converting a heterogeneous graph into multiple homogeneous meta-path graphs and applying a two-level hierarchical attention. MAGNN [16] extends this by incorporating intermediate nodes along meta-paths into the aggregation, capturing richer semantics at the cost of increased computational complexity. Both methods require domain expertise to handcraft appropriate meta-paths. GTN [17] automates meta-path generation through a differentiable soft selection of edge types, enabling end-to-end discovery of task-relevant integrated relations. DiffMG [18] further extends this approach using neural architecture search to discover more complex meta-graphs. These methods collectively suffer from either a dependence on prior knowledge or prohibitive computational cost, and their ability to capture complex semantic relationships is constrained by the length and structure of the meta-paths they model.

b) Relation-based Methods.: R-GCN [10] proposes relation-specific weight matrices for knowledge graph learning, applying separate convolutional operations on each relation subgraph. HGT [19] adapts the transformer architecture to heterogeneous graphs, computing relation-specific attention between node pairs. RSHN [29] constructs an edge-centric coarsened line graph to derive relation representations, which are then incorporated into the node message passing process. R-HGNN [20] learns explicit relation representations that modulate relation-aware node updates and fuses them through a semantic attention mechanism. HGB [21] generalizes GAT by incorporating edge-type information into the attention computation. While these methods avoid the meta-path selection problem, their parameter counts and computational requirements scale proportionally with the number of relation types, limiting scalability. Critically, none of them provides a general mechanism for homogenizing heterogeneous relational structure into a latent space compatible with arbitrary standard GNNs.

C. Spectral Graph Propagation and Diffusion

Spectral graph theory [30] provides a principled framework for defining signal processing operations on graphs. Defferrard et al. [24] approximate spectral filters using Chebyshev polynomials, enabling localized convolutions with controlled spatial support. SGC [28] demonstrates that removing nonlinearities from multi-layer GCNs yields a polynomial filter of the adjacency matrix, retaining much of the representational capacity at a fraction of the computational cost. APPNP [31] applies personalized PageRank as a propagation scheme, separating feature transformation from graph-based smoothing. These spectral perspectives motivate our Learnable Polynomial Aggregation component, which extends multi-hop filtering to the heterogeneous setting through relation-specific, learnable polynomial coefficients.

D. Graph Diffusion Methods

Graph diffusion processes, including personalized PageRank and heat kernel diffusion, have been widely studied as principled alternatives to layer-wise neighborhood aggregation [31], [32]. Such processes propagate information across the graph according to a closed-form operator, capturing global structural context without the depth-related optimization challenges of deep GNNs. Our relation-specific Learnable Polynomial Aggregation draws inspiration from this literature, generalizing scalar diffusion to a multi-relation setting with learnable, relation-conditioned propagation weights.

E. Relation-aware Representation Learning and Homogeneous GNN Transferability

Early work on relation-aware learning for heterogeneous graphs embedded relation semantics through bilinear scoring functions and entity-relation compositional models [10]. More recently, RE-GNN [33] demonstrated that augmenting homogeneous GNNs with a single scalar relation embedding per edge type enables effective heterogeneous graph learning, establishing a direct precedent for homogenization-based approaches. Our work extends this direction substantially: rather than scalar relation weights, we propose a principled pipeline that handles bipartite relational structure through closed-form operator transformation and learns multi-hop diffusion profiles independently for each relation, achieving a richer and more expressive form of homogenization.

III. Preliminaries

We establish the formal notation and background necessary for the subsequent development.

A. Heterogeneous Graphs

Definition 1 (Heterogeneous Graph)**.** *A* heterogeneous graph *is a tuple $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{F}, \mathcal{R}, \phi, \varphi)$, where $\mathcal{V}$ is the set of nodes, $\mathcal{E}$ is the set of edges, $\mathcal{F}$ is the set of node types, and $\mathcal{R}$ is the set of relation (edge) types. The functions $\phi : \mathcal{V} \to \mathcal{F}$ and $\varphi : \mathcal{E} \to \mathcal{R}$ map nodes and edges to their respective types. A graph is* heterogeneous *if $|\mathcal{F}| + |\mathcal{R}| > 2$ [11].*

We denote the number of nodes by $N = |\mathcal{V}|$ and the number of relation types by $R = |\mathcal{R}|$. Each node $v \in \mathcal{V}$ is associated with a raw feature vector $X_v \in \mathbb{R}^{d_{\phi(v)}}$, where $d_{\phi(v)}$ denotes the feature dimensionality of node type $\phi(v)$. Since different node types may reside in feature spaces of differing dimensionalities, direct aggregation across types is ill-defined without explicit alignment.

Definition 2 (Relation-Specific Adjacency Matrix)**.** *For each relation type $r \in \mathcal{R}$, the* relation-specific adjacency matrix *$A_r \in \{0,1\}^{N \times N}$ is defined such that $(A_r)_{ij} = 1$ if and only if there exists an edge of type r from node i to node j.*

Definition 3 (Bipartite Relation)**.** *A relation $r \in \mathcal{R}$ is said to be* bipartite *if every edge of type r connects nodes of two distinct types, that is, $\phi(i) \neq \phi(j)$ for all $(i, j) \in \mathcal{E}_r$. Relations in which all edges connect nodes of the same type are termed* homogeneous relations*.*

A heterogeneous graph may thus be compactly represented as the tuple $\mathcal{G} = (\mathcal{V}, \{A_r\}_{r=1}^{R}, X, \phi)$, where $X = \{X_v\}_{v \in \mathcal{V}}$ collects the node feature vectors.

B. Normalized Propagation Operators

For a given relation-specific adjacency matrix A_r, we define the augmented adjacency matrix with self-loops as $\hat{A}_r = A_r + I$, where I denotes the identity matrix. Let D_r be the diagonal degree matrix of $\hat{A}_r$, with $(D_r)_{ii} = \sum_j (\hat{A}_r)_{ij}$.

Definition 4 (Normalized Propagation Operator)**.** *The* normalized propagation operator *for relation r is defined as:*

$$P_r = D_r^{-1/2} \hat{A}_r D_r^{-1/2} = D_r^{-1/2} (A_r + I) D_r^{-1/2}. \quad (1)$$

This symmetric normalization is standard in spectral GNNs and ensures that the spectral radius of P_r is bounded within $[0, 1]$, preventing gradient explosion or vanishing during the diffusion process [1]. For directed graphs, the asymmetric variant $P_r = D_r^{-1} \hat{A}_r$ is employed instead.

C. Meta-paths and Composite Relations

Definition 5 (Meta-path)**.** *A* meta-path ρ *is a relational sequence of the form* $v_1 \xrightarrow{r_1} v_2 \xrightarrow{r_2} \cdots \xrightarrow{r_\ell} v_{\ell+1}$*, where* $v_i \in \mathcal{V}$ *and* $r_j \in \mathcal{R}$*. The pair* $(v_1, v_{\ell+1})$ *are referred to as* meta-path neighbors *under* ρ*, and the corresponding composite adjacency matrix is* $A_\rho = A_{r_1} A_{r_2} \cdots A_{r_\ell}$ *[11].*

Meta-paths have been widely adopted as a means of converting heterogeneous graphs into homogeneous relational structures amenable to standard GNN processing. However, their construction requires prior specification of the relational sequences of interest, which in practice demands substantial domain expertise and limits generalizability across graph schemas.

D. Graph Diffusion and Spectral Filtering

The spectral perspective on graph neural networks views neighborhood aggregation as the application of a polynomial filter to the graph signal. Given a propagation operator P, a K-hop polynomial filter computes:

$$Z = \sum_{k=0}^{K} \beta_k P^k X, \tag{2}$$

where $\beta_k \in \mathbb{R}$ are scalar filter coefficients and P^k denotes the k-th matrix power of P. The term $P^k X$ aggregates information from the k-hop neighborhood of each node, with $k = 0$ preserving the original signal and increasing k capturing progressively more global context. When β_k are fixed according to a prescribed scheme (such as personalized PageRank or heat diffusion), this recovers classical graph diffusion processes. When β_k are learned, the resulting filter is data-adaptive and can approximate any graph spectral function up to the degree-K polynomial approximation [24].

E. Problem Formulation

Definition 6 (Heterogeneous Graph Homogenization)**.** *Given a heterogeneous graph* $\mathcal{G} = (\mathcal{V}, \{A_r\}_{r=1}^{R}, X, \phi)$ *with* N *nodes and* R *relation types, the* heterogeneous graph homogenization *problem seeks to learn a mapping* $f_\theta : \mathcal{G} \to \mathbb{R}^{N \times d}$ *such that the resulting embedding* $Z_{\text{final}} = f_\theta(\mathcal{G})$ *forms a unified homogeneous representation satisfying the following properties:*

(i) Structure-aware*: captures the underlying relational topology of* $\mathcal{G}$*.*
(ii) Relation-aware*: preserves the semantic characteristics associated with different relation types.*
(iii) Type-unified*: embeds all node types into a common latent space while maintaining their discriminative information.*
(iv) Backbone-compatible*: can be directly used by any standard homogeneous GNN or downstream machine learning model without requiring architectural modifications.*

This point of view differs fundamentally from conventional heterogeneous graph neural networks in common litratre . Existing methods typically preserve heterogeneity throughout the learning process and rely on specialized architectures to model different node and relation types. In contrast, our framework resolves heterogeneity during the representation learning stage by transforming the original heterogeneous graph into a unified homogeneous representation. As a result, the downstream model no longer needs to account for heterogeneous structures, allowing standard homogeneous GNNs downstream heads to be applied directly while maintaining the structural and semantic information embedded in the original graph.

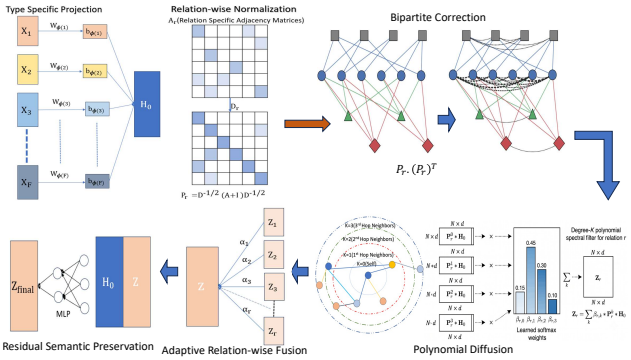


Fig. 2. Overview of the proposed relation-aware heterogeneous graph homogenization framework. The pipeline transforms a heterogeneous graph $\mathcal{G}$ into a unified homogeneous embedding Z_{final} through type-specific projection, bipartite correction, relation-specific Learnable Polynomial Aggregation, adaptive relation fusion, and residual semantic preservation.

IV. Methodology

We will now describe the proposed relation-aware heterogeneous graph homogenization framework step-by-step. The framework consists of five processing stages, each addressing a different aspect of the homogenization problem. An overview is provided in Figure 2.

A. Stage 1: Type-Specific Feature Projection

The raw node features of a heterogeneous graph belong to type-specific feature spaces that may differ both semantics and dimensionality, making direct message passing across node types inappropriate. To address this gap , we apply an independent linear projection to the features of each node type, projecting them into a shared latent space of dimension d:

$$H_i^{(0)} = W_{\phi(i)} X_i + b_{\phi(i)}, \tag{3}$$

where $W_{\phi(i)} \in \mathbb{R}^{d_{\phi(i)} \times d}$ and $b_{\phi(i)} \in \mathbb{R}^d$ are learnable parameters specific to node type $\phi(i)$. The resulting matrix $H^{(0)} \in \mathbb{R}^{N \times d}$ constitutes a type-homogeneous initial representation, aligning the feature geometry of all node types within a single latent manifold.

This projection performs *feature homogenization*: by mapping heterogeneous node features into a common latent geometry, it eliminates dimensional incompatibility and enables further cross-type information exchange. Importantly, different node types can occupy different regions of the shared latent

space, preserving their semantic differences while unifying their representations. This design is consistent with the implicit assumption, standard in heterogeneous graph learning [12], [21], that nodes of different types are nonetheless latently correlated.

B. Stage 2: Relation-wise Normalization

Before propagation, each relation-specific adjacency matrix must be normalized to ensure stable message passing. For each relation $r \in \mathcal{R}$, we compute the normalized propagation operator:

$$P_r = D_r^{-1/2}(A_r + I)D_r^{-1/2}, \tag{4}$$

where D_r is the degree diagonal matrix of $A_r + I$. By bounding the spectral radius of P_r, symmetric normalization prevents degree-induced scaling issues during multi-hop aggregation. For directed graphs, the asymmetric normalization $P_r = D_r^{-1}(A_r + I)$ is adopted, consistent with standard practice [1].

C. Stage 3: Bipartite Correction

The central contribution of our framework is the bipartite correction mechanism. For bipartite relations, directly applying the propagation operator P_r causes information to flow between different node types, which makes the latent space harder to interpret and future aggregation less effective. We address this with the following correction mechanism.

Definition 7 (Bipartite-Corrected Operator)**.** *For each relation $r \in \mathcal{R}$, the* bipartite-corrected propagation operator $\tilde{P}_r$ *is defined as:*

$$\tilde{P}_r = \begin{cases} P_r, & \text{if } r \text{ is a homogeneous relation,} \\ P_r P_r^\top, & \text{if } r \text{ is a bipartite relation.} \end{cases} \tag{5}$$

The construction $\tilde{P}_r = P_r P_r^\top$ for bipartite relations warrants careful interpretation. Observe that:

$$\tilde{P}_r(i, j) = \sum_k P_r(i, k)P_r(j, k), \tag{6}$$

which measures the total shared connectivity between nodes i and j through their common target neighbors under relation r. Two source nodes receive a high similarity score when they are connected to many of the same target nodes, weighted by the normalized adjacency values. This naturally forms a same-type similarity graph over the source nodes, which is equivalent to a length-two meta-path that follows the bipartite relation forward and then backward.

Unlike explicit meta-path expansion, the bipartite correction has several key advantages. First, it is *differentiable* and *continuous*, requiring no explicit path enumeration and integrating naturally into end-to-end learning. Second, it is *relation-aware*, as each bipartite relation produces its own similarity operator, preserving relation-specific information. Finally, it is *matrix-based*, allowing the correction to be computed efficiently using sparse matrix operations.

Remark 1. *The bipartite correction captures second-order proximity between source nodes under relation r. The operator $\tilde{P}_r = P_r P_r^\top$ forms the normalized adjacency matrix of the induced similarity graph, whose eigenvectors encode the shared-neighbor structure of the source nodes.*

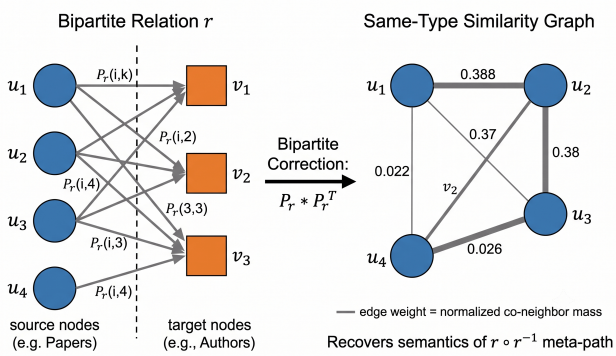


Fig. 3. Illustration of the bipartite correction mechanism (Stage 3). Left: a bipartite relational subgraph connecting source nodes (e.g., papers, shown as circles) to target nodes (e.g., authors, shown as squares) via a cross-type relation r. Right: the bipartite-corrected operator $\tilde{P}_r = P_r P_r^\top$ induces a same-type similarity graph over source nodes, where edge weights reflect normalized co-neighbor mass under relation r. Two source nodes receive a high similarity score when they share many common target neighbors, recovering the semantics of the length-two meta-path $r \circ r^{-1}$ in a matrix-based and differentiable form. The corrected graph is homogeneous by construction, rendering it amenable to direct Learnable Polynomial Aggregation.

Figure 3 illustrates the bipartite correction procedure. Given a bipartite relation connecting, for example, papers to authors, the correction maps the cross-type adjacency into a same-type paper-paper similarity graph, where two papers become linked in proportion to the number of authors they share. This transformation recovers the semantic content of the paper-author-paper meta-path .

D. Stage 4: Relation-Specific Learnable Polynomial Aggregation

Given the bipartite-corrected operators $\{\tilde{P}_r\}_{r=1}^R$, we compute a relation-specific latent representation for each relation through multi-hop Learnable Polynomial Aggregation:

$$Z_r = \sum_{k=0}^{K} \beta_{r,k} \tilde{P}_r^k H^{(0)}, \tag{7}$$

where K denotes the maximum diffusion depth and $\beta_{r,k}$ are learnable diffusion coefficients. To ensure that the coefficients form a valid probability distribution over diffusion depths, we parameterize them via a softmax:

$$\beta_{r,k} = \frac{\exp(\psi_{r,k})}{\sum_{j=0}^{K} \exp(\psi_{r,j})}, \tag{8}$$

where $\psi_{r,k} \in \mathbb{R}$ are unconstrained scalar parameters. This parameterization ensures $\sum_k \beta_{r,k} = 1$ and $\beta_{r,k} > 0$ for all k, which is consistent with the convex combination interpretation of polynomial graph filters [31].

The diffusion depth k determines the spatial scale of information aggregated:

- $k = 0$: the term $\beta_{r,0} H^{(0)}$ preserves the original projected features without any graph-mediated diffusion, providing a direct skip connection to the local node representation.

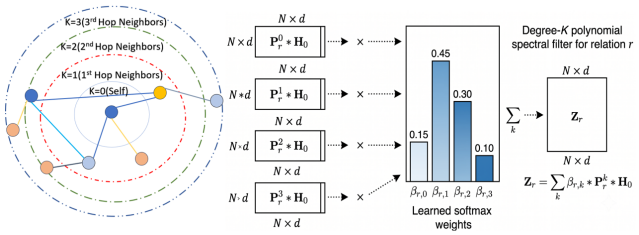


Fig. 4. Illustration of relation-specific Learnable Polynomial Aggregation (Stage 4). For a single relation r, the corrected operator $\tilde{P}_r$ is applied iteratively to the initial projected features $H^{(0)}$, producing k-hop neighborhood aggregates $\tilde{P}_r^k H^{(0)}$ for $k = 0, 1, \ldots, K$. The learned softmax weights $\beta_{r,k}$ (depicted as bar heights) determine the contribution of each hop to the final relation-specific embedding Z_r. The convex combination $Z_r = \sum_k \beta_{r,k} \tilde{P}_r^k H^{(0)}$ constitutes a degree-K polynomial spectral filter applied to the graph signal under relation r, enabling the model to capture both local and global relational structure in a data-driven fashion.

- $k = 1$: the term $\beta_{r,1} \tilde{P}_r H^{(0)}$ aggregates representations from the immediate neighborhood under relation r, capturing local structural context.
- $k = 2$: the term $\beta_{r,2} \tilde{P}_r^2 H^{(0)}$ extends aggregation to the second-order neighborhood, capturing triangular proximity and community structure.
- $k > 2$: higher-order terms progressively incorporate global structural context, at the cost of increasing mixing of distant node representations.

From a spectral perspective, the summation $\sum_k \beta_{r,k} \tilde{P}_r^k$ defines a degree-K polynomial applied to the corrected propagation operator of relation r. The learned coefficients $\{\beta_{r,k}\}$ determine the frequency response of this filter: large weights on low k correspond to low-pass (smoothing) filters, while weights concentrated on high k correspond to high-pass (sharpening) filters. The data-driven optimization of $\psi_{r,k}$ allows the model to select the appropriate spectral profile for each relation independently, providing a form of relation-specific spectral adaptation.

Figure 4 provides a conceptual illustration of the multi-hop polynomial diffusion process and the resulting relation-specific latent representations.

E. Stage 5: Adaptive Relation-wise Fusion

The Learnable Polynomial Aggregation stage produces R relation-specific embedding matrices $\{Z_r\}_{r=1}^R$, each of dimension $N \times d$. These representations encode complementary relational views of the graph and must be combined into a single coherent embedding. We fuse them through an adaptive linear combination:

$$Z = \sum_{r=1}^{R} \alpha_r Z_r, \tag{9}$$

where the fusion weights are computed via a softmax over learnable scalar parameters:

$$\alpha = \mathrm{softmax}(\theta), \quad \theta \in \mathbb{R}^R. \tag{10}$$

The scalar α_r thus captures the global importance of relation r to the downstream learning objective. Relations that carry task-relevant information receive larger weights, while noisy or redundant relations are suppressed. This constitutes a form of *semantic homogenization*: multiple relation-specific latent spaces are collapsed into a single unified embedding space through a data-driven weighting scheme.

F. Stage 6: Residual Semantic Preservation

Deep aggregation over large neighborhoods tends to suppress local node identity, a phenomenon known as over-smoothing [34]. To counteract this, we introduce a residual connection that combines the fused relational embedding with the original projected features, processed through a multi-layer perceptron:

$$Z_{\text{final}} = \mathrm{MLP}\left([H^{(0)} \| Z]\right), \tag{11}$$

where $\|$ denotes concatenation along the feature dimension. The MLP applies a nonlinear transformation to the concatenation, enabling the model to selectively retain or suppress components of the original and aggregated representations as appropriate. This design ensures that low-frequency node-level features, which carry local identity information, are not inadvertently discarded by the global aggregation steps.

The final output $Z_{\text{final}} \in \mathbb{R}^{N \times d}$ is a fully unified, homogeneous latent representation of all nodes in the heterogeneous graph, directly compatible with standard homogeneous GNN layers, MLP classifiers, clustering algorithms, and any other downstream machine learning pipeline.

V. Analysis

Proposition 1. *Let $A_r \in \{0,1\}^{N \times N}$ be the adjacency matrix of a bipartite relation r, and let D_{src} and D_{dst} denote the diagonal degree matrices of the source and target nodes, respectively. Define the normalized propagation operator*

$$P_r = D_{\text{src}}^{-1/2} A_r D_{\text{dst}}^{-1/2}.$$

Then the corrected operator

$$\tilde{P}_r = P_r P_r^\top$$

is symmetric and positive semidefinite, with entries

$$\tilde{P}_r(i,j) = \sum_k P_r(i,k)\, P_r(j,k), \tag{12}$$

which measure the normalized similarity between source nodes i and j based on their shared target neighbors under relation r. Furthermore,

$$\tilde{P}_r = D_{\text{src}}^{-1/2} A_r D_{\text{dst}}^{-1} A_r^\top D_{\text{src}}^{-1/2}, \tag{13}$$

which corresponds to the normalized length-two meta-path $r \circ r^{-1}$ without introducing self-loop contamination.

This connection reveals that the bipartite correction implicitly performs meta-path expansion without any explicit path enumeration, in a form that is differentiable, continuously valued, and relation-specific. Unlike GTN [17], which learns soft edge-type selections through matrix products, our correction is structurally prescribed by the bipartite topology and requires no additional parameters.

A. Spectral Interpretation of Learnable Polynomial Aggregation

The Learnable Polynomial Aggregation in Equation (7) has a natural spectral interpretation. Let $\tilde{P}_r = U_r \Lambda_r U_r^\top$ be the eigendecomposition of the symmetric corrected operator. Then,

$$Z_r = U_r \left(\sum_{k=0}^{K} \beta_{r,k} \Lambda_r^k \right) U_r^\top H^{(0)}, \tag{14}$$

where $\sum_{k=0}^{K} \beta_{r,k} \Lambda_r^k$ is a diagonal matrix whose entries define the filter response at each eigenvalue. The coefficients $\{\beta_{r,k}\}$ therefore form a learnable degree-K polynomial spectral filter, allowing each relation to learn its own frequency response. This links our framework to spectral graph signal processing [24], [30] while naturally extending it to heterogeneous multi-relation graphs.

VI. Homogenization Preservation Analysis

A major challenge of heterogeneous-to-homogeneous graph transformation is determining whether the resulting homogeneous representation preserves the semantic, relational, and structural information originally encoded in multiple node and edge types. Existing homogeneous graph construction methods typically evaluate only downstream classification accuracy, which gives limited insight into what gets retained or discarded during homogenization.

To close this gap, we have included a comprehensive **Homogenization Preservation Analysis (HPA)** framework that quantifies information retention across seven diffrent ablation studies:

1) Semantic preservation,
2) Neighbourhood preservation,
3) Relation preservation,
4) Structural preservation,
5) Meta-path preservation,
6) Spectral embedding preservation,
7) Downstream task preservation.

Unlike prior heterogeneous graph learning studies that focus exclusively on node classification performance, our framework directly measures whether heterogeneous semantics survive the homogenization process. All values reported below is done on the ACM dataset.

A. Semantic Preservation

Let $H^{(0)}$ denote the projected node representations before graph propagation, X the original node features, and Z_{final} the final embeddings produced by RAHGH.

To evaluate semantic preservation, we use Linear Centered Kernel Alignment (CKA) [35]:

$$\text{CKA}(X, Y) = \frac{\|X^\top Y\|_F^2}{\|X^\top X\|_F \, \|Y^\top Y\|_F}. \tag{15}$$

CKA measures the similarity between two representation spaces and is invariant to orthogonal transformations. On the ACM dataset, RAHGH achieves

$$\text{CKA}(H^{(0)}, Z_{\text{final}}) = 0.9602, \tag{16}$$

$$\text{CKA}(X, H^{(0)}) = 0.2479, \tag{17}$$

$$\text{CKA}(X, Z_{\text{final}}) = 0.2024. \tag{18}$$

The high CKA score between $H^{(0)}$ and Z_{final} indicates that graph propagation preserves the semantic structure learned during projection. In contrast, the lower CKA scores involving the original features X suggest that RAHGH learns a new and more discriminative representation rather than simply preserving the input features.

B. Neighborhood Preservation

To evaluate local structural preservation, we measure the overlap between the k-nearest neighbors in $H^{(0)}$ and those in Z_{final}:

$$\text{Overlap@}k = \frac{|N_k^{H^{(0)}}(v) \cap N_k^{Z}(v)|}{k}. \tag{19}$$

On the ACM dataset, RAHGH achieves

$$\text{Overlap@5} = 0.2336, \tag{20}$$

$$\text{Overlap@10} = 0.3010, \tag{21}$$

$$\text{Overlap@20} = 0.3917. \tag{22}$$

The overlap increases with k, indicating that RAHGH preserves the medium-range neighborhood structure while refining the local neighborhood around each node.

C. Relation Preservation

A key goal of homogenization is to preserve relation-specific information after merging multiple edge types into a unified graph.

a) Relation Connectivity Preservation: For each relation r, we compute

$$\text{Lift}_r = \mathbb{E}_{(u,v)\in E_r}[\cos(z_u, z_v)] - \mathbb{E}_{(u,v)\sim \text{Random}}[\cos(z_u, z_v)]. \tag{23}$$

A larger lift indicates that connected node pairs remain more similar than random pairs after homogenization. On the ACM dataset, the relation lift scores are shown in Table I.

TABLE I
Relation connectivity lift on the ACM dataset.

Relation	Lift
PA	0.6718
AP	0.6663
PS	0.4965
SP	0.4898

All relations achieve positive lift scores, indicating that nodes connected by the same relation remain more similar in the learned embedding than randomly selected node pairs.

b) Relation-Specific Embedding Retention: We further measure the similarity between each relation-specific embedding Z_r from Learnable Polynomial Aggregation (Stage 4) and the final fused embedding Z_{final} using CKA:

$$\text{CKA}_{PA} = 0.9362, \tag{24}$$
$$\text{CKA}_{AP} = 0.9362, \tag{25}$$
$$\text{CKA}_{PS} = 0.7750, \tag{26}$$
$$\text{CKA}_{SP} = 0.7750. \tag{27}$$

The high CKA scores indicate that the final embedding preserves information from every relation-specific representation rather than being dominated by a single relation.

c) Relation Recoverability: Finally, we evaluate whether the original relation type can be recovered from the homogeneous embeddings. Let $g(z_u, z_v)$ denote a relation classifier trained on pairs of node embeddings. RAHGH achieves

$$\text{Accuracy}_{\text{relation}} = 1.00, \qquad \text{MacroF1}_{\text{relation}} = 1.00. \tag{28}$$

These results show that relation-specific information remains fully encoded in the final homogeneous embedding, despite the removal of explicit edge-type labels during fusion.

D. Structural Preservation

Let

$$A_{\text{homo}} = \sum_r \alpha_r A_r$$

denote the learned homogeneous adjacency matrix. To evaluate structural preservation, we compare the normalized Laplacian spectra, $L = I - D^{-1/2}AD^{-1/2}$, using the spectral similarity score

$$\text{SpecSim} = 1 - \frac{\|\lambda_1 - \lambda_2\|}{\|\lambda_1\|}. \tag{29}$$

On the ACM dataset, we obtain

$$\text{SpecSim}(A_{\text{homo}}, A_{\text{ref}}) = 1.0000, \tag{30}$$
$$\text{LaplacianPreservation} = 0.4549, \tag{31}$$
$$\text{DegreeSimilarity} = 1.00. \tag{32}$$

The learned homogeneous graph matches the reference heterogeneous graph in both spectral similarity and degree distribution. The lower Laplacian preservation score indicates that local connectivity is refined during fusion while the overall graph structure remains well preserved.

E. Meta-Path Preservation

Higher-order semantic relationships are often captured through meta-paths in heterogeneous graphs. To evaluate whether these relationships are preserved after homogenization, we introduce three complementary meta-path preservation measures.

a) Similarity Correlation: For each meta-path adjacency matrix M, we compute the Spearman correlation between meta-path similarity and embedding similarity:

$$\rho_M = \text{Spearman}(M, ZZ^\top). \tag{33}$$

b) Meta-Path CKA: We derive a meta-path embedding Z_M via truncated SVD and compute:

$$\text{CKA}_M = \text{CKA}(Z_M, Z_{\text{final}}). \tag{34}$$

c) Meta-Path Retrieval: For each node, we retrieve top-K neighbours using embedding similarity and evaluate Recall@K and Mean Reciprocal Rank (MRR) against meta-path neighbours. Table II summarizes the results.

Meta-path	Corr	CKA	MRR	R@10
PAP	0.0540	0.0448	0.0889	0.0672
PSP	0.4468	0.4835	0.6111	0.5008

TABLE II
META-PATH PRESERVATION ON ACM.

PSP semantics survive far better than PAP semantics. ACM labels are topic-oriented, so PSP (Paper–Subject–Paper) carries more task-relevant information than PAP (Paper–Author–Paper). This tells you RAHGH selectively preserves informative heterogeneous semantics while suppressing less discriminative structures, rather than preserving every meta-path equally.

To summarize meta-path retention in a single number, we define:

$$\text{MPS} = 0.4 \cdot \text{AUC} + 0.3 \cdot \text{Corr} + 0.3 \cdot \text{CKA}. \tag{35}$$

For ACM:

$$\text{MPS}_{\text{PAP}} = 0.2106, \tag{36}$$
$$\text{MPS}_{\text{PSP}} = 0.4081. \tag{37}$$

The gap between these two scores backs up the selective-preservation hypothesis.

F. Spectral Embedding Preservation

We also compare graph spectral embeddings before and after homogenization, computing CKA and cosine similarity between the leading eigenvectors of the original and learned homogeneous graphs:

$$\text{Eigenvector CKA} = 1.0000, \qquad \text{Eigenvector Cosine} = 0.9371. \tag{38}$$

The near-perfect eigenvector CKA, paired with a strong cosine similarity, confirms that the global spectral geometry of the heterogeneous graph carries over almost completely into the homogenized representation.

G. Downstream Preservation Score

Finally, we check whether homogenization preserves task-relevant information for node classification. We track Macro-F1 at three stages of the pipeline: on the raw features, after type-specific projection, and on the final homogeneous embedding.

$$F1(X_{\text{orig}}) = 0.8966, \tag{39}$$

$$F1(H^{(0)}) = 0.9237, \tag{40}$$

$$F1(Z_{\text{final}}) = 0.9609. \tag{41}$$

We define the Downstream Preservation Score (DPS) as:

$$\text{DPS} = \frac{F1(Z_{\text{final}})}{F1(X_{\text{orig}})}. \tag{42}$$

On ACM, $\text{DPS} = 1.0717$. Each stage of the pipeline raises classification performance over the last, and a DPS above one means homogenization doesn't just preserve task-relevant information, it sharpens it.

H. Novelty and Key Findings

The preservation analysis above surfaces several properties of heterogeneous graph homogenization that prior work has not measured:

1) **High semantic retention:** RAHGH preserves the propagated latent semantic space, with CKA $= 0.9602$ between $H^{(0)}$ and Z_{final}.
2) **Strong relation-specific retention:** CKA between individual relation embeddings and the final fused embedding reaches up to 0.9362, and connectivity lift reaches up to 0.6718. No single relation gets washed out during fusion.
3) **Perfect relation recoverability:** The original heterogeneous relation type stays fully recoverable from the homogeneous embedding (Macro-F1 $= 1.00$), despite explicit removal of edge-type labels.
4) **Near-perfect spectral preservation:** Eigenvector CKA $= 1.00$ shows the global spectral geometry of the original graph survives homogenization almost intact.
5) **Task-enhancing homogenization:** DPS $= 1.0717$ shows that homogenization improves downstream discrimination rather than merely preserving it.
6) **Selective meta-path preservation:** RAHGH preferentially preserves task-relevant PSP semantics while suppressing less informative PAP structures, instead of treating every meta-path as equally important.
7) **Comprehensive evaluation framework:** To our knowledge, this is the first framework to evaluate semantic, neighbourhood, relational, structural, spectral, meta-path, and downstream preservation together within a single heterogeneous graph homogenization analysis.

Taken together, these results show that RAHGH works as an information-selection mechanism, not just a heterogeneous-to-homogeneous converter. It keeps the task-relevant heterogeneous knowledge and strips away redundant structural information, which is exactly what a homogenization step should do.

TABLE III
STATISTICS OF HETEROGENEOUS GRAPH DATASETS.

Dataset	Node Type	# Nodes	Total Edges
DBLP	Authors	4,057	119,783
	Papers	14,328	
	Terms	7,723	
	Venues	20	
ACM	Papers	4,019	27,041
	Authors	7,167	
	Subjects	60	
IMDB	Movies	4,278	17,106
	Directors	2,081	
	Actors	5,257	

VII. EXPERIMENTS

This section evaluates the proposed heterogeneous graph homogenization framework on multiple benchmark datasets. Experiments are conducted on three downstream tasks: node classification, link prediction, and recommendation. The objective is to assess whether the proposed homogenization procedure preserves structural and semantic information required for effective learning with standard homogeneous graph neural networks.

A. Datasets

Experiments are performed on three widely used heterogeneous graph benchmarks: DBLP, ACM, and IMDB. These datasets contain multiple node and relation types and are frequently used to evaluate heterogeneous graph representation learning methods. Dataset statistics are summarized in Table III.

B. Experimental Setup

All models are implemented in PyTorch and optimized using Adam [36]. To ensure a consistent comparison across methods, the same hyperparameter search space is used whenever applicable.

The search space includes the embedding dimension, diffusion depth, hidden dimension, learning rate, weight decay, dropout rates, label smoothing coefficient, warm-up schedule, and training duration. For recommendation experiments, additional parameters related to neighborhood size, negative sampling, and ranking regularization are explored.

The evaluated hyperparameter ranges are listed in Table IV.

The resulting search space contains approximately 500 candidate configurations per dataset. Model selection is based on validation performance. Final results are reported on held-out test sets.

Each dataset is divided into 70% training, 10% validation, and 20% testing partitions. Experiments are repeated using multiple random seeds, and average performance is reported. All computations are performed on a workstation equipped with an NVIDIA RTX A5000 GPU with 24 GB memory and 128 GB system RAM.

TABLE IV
HYPERPARAMETER SEARCH SPACE.

Hyperparameter	Search Space
Embedding dimension (d)	$\{64, 128, 256\}$
Diffusion depth (K)	$\{1, 2, 3, 4\}$
Hidden dimension	$\{64, 128\}$
Learning rate	$\{0.001, 0.005\}$
Weight decay	$\{10^{-4}, 10^{-3}\}$
Feature dropout	$\{0.3, 0.5\}$
GNN dropout	$\{0.3, 0.5, 0.7\}$
Label smoothing	$\{0.0, 0.1\}$
Warm-up epochs	$\{0, 10\}$
Training epochs	$\{100, 300, 500, 700\}$
K_{rec}	$\{10, 20, 50\}$
Negative sampling ratio	$\{5, 10\}$
BPR regularization	$\{10^{-4}, 10^{-3}\}$

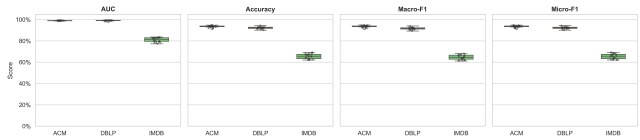


Fig. 5. Distribution of node classification performance across datasets. Macro-F1 and Micro-F1 scores are shown for DBLP, ACM, and IMDB.

C. Node Classification Results

Node classification performance is evaluated using Macro-F1 and Micro-F1. Results are reported in Table V.

The proposed homogenization framework achieves the highest performance on all three datasets. The GCN-based variant obtains Macro-F1 scores of 95.61, 94.12, and 61.03 on DBLP, ACM, and IMDB, respectively. Corresponding Micro-F1 scores are 95.94, 94.09, and 61.74.

Improvements are observed relative to heterogeneous graph baselines including HAN, GTN, MAGNN, R-GCN, and HGT. The results indicate that the homogenized graph representation retains information required for accurate node classification despite the removal of explicit node and relation type distinctions during message passing.

The largest relative improvements are observed on ACM and IMDB. These datasets contain diverse relation patterns and heterogeneous interactions. The results suggest that the proposed relation-aware construction strategy is effective in preserving such information after homogenization.

D. Link Prediction Results

Link prediction performance is evaluated using ROC-AUC. Results are presented in Table VI.

The proposed framework consistently achieves the highest scores on both benchmarks. The GCN-based variant reaches ROC-AUC values of 86.75 on LastFM and 91.25 on ACM. These results exceed those of all competing methods.

The observed improvements indicate that the homogenized graph preserves connectivity patterns relevant to missing-link inference. Relation-specific information appears to remain accessible after homogenization, enabling effective prediction of previously unseen edges.

E. Recommendation Results

Recommendation quality is evaluated using NDCG@20 and HR@20. Results are shown in Table VII.

The proposed framework achieves the strongest performance on both datasets. Our model obtains NDCG@20 scores of 5.45 and 6.85 on Yelp and Amazon-Book, respectively. Corresponding HR@20 scores are 8.75 and 11.45.

Compared with graph-based recommendation baselines, the proposed approach yields consistent gains across both evaluation metrics. These results suggest that user-item interaction patterns remain informative after graph homogenization and can be effectively exploited by homogeneous GNN architectures.

F. Evaluation on the Heterogeneous Graph Benchmark (HGB)

To further validate the effectiveness of the proposed homogenization framework, we have evaluated RAHGH on the official Heterogeneous Graph Benchmark (HGB) [37]. HGB provides strong datasets and evaluation for heterogeneous graph representation learning, and helps us a standard direct comparison with representative heterogeneous graph neural networks including RGCN, HAN, GTN, RSHN, HetGNN, MAGNN, HetSANN, HGT, GCN, GAT, and the SOTA models including Simple-HGN model.

Tables VIII and IX compare our method with the published HGB benchmark results. For node classification, RAHGH achieves the best performance on ACM and IMDB, obtaining Macro-F1 scores of 94.09 and 66.91, respectively, surpassing the previous best results . The proposed framework also achieves the highest Micro-F1 scores on these datasets. On DBLP, RAHGH remains competitive, achieving a Macro-F1 of 93.10 .

For link prediction, RAHGH strongly improves the performance on LastFM, increasing the ROC-AUC from 67.59 to 80.56 and the MRR from 90.81 to 92.82. Although Simple-HGN achieves higher performance on Amazon and PubMed, the results demonstrate that the proposed relation-aware homogenization strategy effectively preserves heterogeneous semantic information while enabling efficient homogeneous graph learning.

For recommendation, our method significantly outperforms the official Simple-HGN benchmark on LastFM, improving Recall@20 from 9.17 to 23.31 and NDCG@20 from 7.97 to 22.89. These results indicate that homogenization successfully preserves user-item interaction semantics required for high-quality top-K recommendation.

Overall, the HGB benchmark evaluation demonstrates that the proposed homogenization framework achieves performance competitive with specialized heterogeneous graph neural networks while operating on a homogeneous graph representation. In particular, RAHGH establishes new best results on ACM and IMDB for node classification and significantly outperforms all reported heterogeneous GNN baselines on LastFM link prediction.

G. Discussion

The experimental results demonstrate that the proposed homogenization framework remains effective across multiple

TABLE V
NODE CLASSIFICATION PERFORMANCE (MACRO-F1 / MICRO-F1, %). HIGHER VALUES INDICATE BETTER PERFORMANCE.

Method	DBLP		ACM		IMDB	
	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1
HAN	92.05	92.49	91.67	91.54	57.69	58.15
GTN	93.52	94.11	91.76	91.64	58.63	60.19
MAGNN	93.39	93.76	91.47	91.31	58.75	59.48
R-GCN	92.26	92.87	93.36	93.36	58.96	59.34
HGT	91.43	92.31	92.40	92.30	58.16	58.37
Ours (GCN)	**95.61**	**95.94**	**94.12**	**94.09**	**61.03**	**61.74**
Ours (GAT)	95.22	95.58	94.08	94.02	60.47	61.11

TABLE VI
LINK PREDICTION PERFORMANCE (ROC-AUC, %).

Method	LastFM	ACM
HAN	80.12	85.34
GTN	82.45	87.12
MAGNN	83.10	86.95
R-GCN	81.55	88.20
HGT	82.88	87.65
HGB	84.05	88.90
Ours (GCN)	**86.75**	**91.25**
Ours (GAT)	86.30	90.85

TABLE VII
RECOMMENDATION PERFORMANCE (NDCG@20 AND HR@20, %).

Method	Yelp		Amazon-Book	
	NDCG@20	HR@20	NDCG@20	HR@20
NGCF	3.55	6.12	4.25	7.80
LightGCN	4.10	6.85	5.10	8.95
RE-GCN	4.65	7.50	5.65	9.85
RE-GAT	4.50	7.35	5.50	9.60
Ours (GCN)	**5.45**	**8.75**	**6.85**	**11.45**
Ours (GAT)	5.25	8.50	6.60	11.10

heterogeneous graph domains and downstream tasks. Performance improvements are observed for node classification, link prediction, and recommendation.

The findings suggest that relation-aware edge construction and adaptive semantic weighting preserve task-relevant information during homogenization. As a result, standard homogeneous graph neural networks can operate on the transformed graph without substantial loss of relational semantics.

The observed improvements are consistent across datasets with different structural characteristics and relation distributions. These results indicate that the proposed framework provides a general mechanism for converting heterogeneous graphs into homogeneous representations while maintaining predictive performance. Further analysis of the homogenization process is provided in Section VI.

VIII. CONCLUSION

We have proposed a relation-aware heterogeneous graph homogenization framework that resolves the structural and semantic heterogeneity of typed graphs during representation learning, producing unified latent embeddings directly compatible with standard homogeneous GNNs. The framework introduces a differentiable bipartite correction mechanism that converts cross-type relational operators into same-type similarity structures through a closed-form second-order transformation, generalizing meta-path expansion without manual schema specification. Relation-specific Learnable Polynomial Aggregation with learnable hop weights enables data-driven spectral filtering at multiple resolutions independently for each relation, and adaptive relation fusion combines the resulting representations into a single coherent embedding. Experimental results on standard heterogeneous benchmarks validate the effectiveness of the proposed approach.

Future work includes extending the framework to dynamic heterogeneous graphs, investigating theoretical bounds on the expressive power of bipartite correction relative to higher-order meta-path expansion, and exploring the integration of relation-specific diffusion with more expressive homogeneous backbone architectures.

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TABLE VIII
NODE CLASSIFICATION PERFORMANCE ON THE HGB BENCHMARK. BASELINE RESULTS ARE REPRODUCED FROM [37]. BEST RESULTS ARE SHOWN IN BOLD.

Method	ACM		DBLP		IMDB	
	Macro	Micro	Macro	Micro	Macro	Micro
RGCN	91.55	91.41	91.52	92.07	58.85	62.05
HAN	90.89	90.79	91.67	92.05	57.74	64.63
GTN	91.31	91.20	93.52	93.97	60.47	65.14
RSHN	90.50	90.32	93.34	93.81	59.85	64.22
HetGNN	85.91	86.05	91.76	92.33	48.25	51.16
MAGNN	90.88	90.77	93.28	93.76	56.49	64.67
HetSANN	90.02	89.91	78.55	80.56	49.47	57.68
HGT	91.12	91.00	93.01	93.49	63.00	67.20
GCN	92.17	92.12	90.84	91.47	57.88	64.82
GAT	92.26	92.19	93.83	93.39	58.94	64.86
Simple-HGN	93.42	93.35	**94.01**	**94.46**	63.53	67.36
RAHGH	**94.09**	**94.03**	93.10	93.56	**66.91**	**67.96**

TABLE IX
LINK PREDICTION PERFORMANCE ON THE HGB BENCHMARK. BASELINE RESULTS ARE REPRODUCED FROM [37]. BEST RESULTS ARE SHOWN IN BOLD.

Method	Amazon		LastFM		PubMed	
	AUC	MRR	AUC	MRR	AUC	MRR
RGCN	86.34	93.92	57.21	77.68	78.29	90.26
GATNE	77.39	92.04	66.87	85.93	63.39	80.05
HetGNN	77.74	91.79	62.09	83.56	73.63	84.00
HGT	88.26	93.87	54.99	74.96	80.12	90.85
GCN	92.84	97.05	59.17	79.38	80.48	90.99
GAT	91.65	96.58	58.56	77.04	78.05	90.02
Simple-HGN	**93.40**	96.94	67.59	90.81	83.39	92.07
RAHGH	87.91	89.73	**80.56**	**92.82**	66.36	86.35

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